

Ishraque Zaman Borshon

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SUMMARY

PhD candidate (AME, University of Arizona) with ~7 years research and ~3 years industry experience in applied ML, computer vision, and scientific/materials engineering. First-author Q1 publications and contributor to production-grade open-source scientific software.

EDUCATION

Ph.D., Aerospace & Mechanical Engineering, University of Arizona

Expected July 2026

GPA: 3.5/4.0 | Dissertation: *Physics-Informed Deep Learning for Atomic-Scale Characterization of Electrochemical Energy Materials*

M.Tech, Energy Systems Engineering, IIT Bombay

2019 – 2021

B.Tech, Mechanical Engineering, NIT Kurukshetra

2012 – 2016

TECHNICAL SKILLS

ML & DL: Computer vision, ViT, U-Net, Mask R-CNN, segmentation, object detection, physics-informed ML, transfer learning, uncertainty quantification, active learning, VAEs, RL

Agentic AI & LLMs: Prompt engineering, LLM fine-tuning, RAG, LangGraph, Claude Agent SDK, multi-agent orchestration

Frameworks: PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, Streamlit, RLLib

Programming & Tools: Python, C, Julia, VB.NET, C#, MATLAB, SQL, Git, Linux, Docker (exposure), AWS/Azure/GCP (exposure), SLURM, HPC

Microscopy & Simulation: SEM/STEM/TEM analysis, PyQSTEM multislice, DFT, ANSYS Fluent (CFD), COMSOL

EXPERIENCE

Doctoral Researcher, University of Arizona

Aug 2023 – Present

- Developed ViT, U-Net, and Mask R-CNN models for atomic-column segmentation, defect detection, and interface classification in STEM/TEM micrographs of battery and energy materials.
- Built physics-informed, simulation-to-experiment pipelines (PyQSTEM multislice) for robust inference on noisy, low-contrast experimental data; predicted column heights and composition in high-entropy oxides.
- Quantified SEI microstructures in lithium-based batteries to link defects and interfaces to degradation, state-of-health, and cycle life; trained models on GPU-accelerated HPC clusters via SLURM.
- Implemented active-learning pipelines and explored VAEs to generate atomic-scale training data in low-data regimes; applied Bayesian uncertainty quantification for robust inference.
- Developed agentic AI and GenAI features for simulation workflows using LLM orchestration, RAG, and prompt engineering.

Graduate Research Assistant, Oklahoma State University

Aug 2021 – Jul 2023

- Co-developed GLHEPro 5.1 and open-source GHEDesigner (NREL SWR-23-33, with OSU and NREL); scaled borefield design from 308 to 34,321 configurations in VB.NET; supported hundreds of engineers.
- Contributed to an ORNL techno-economic modeling report for geothermal heat-pump applications (ORNL/TM-2023/2933); performed sensitivity analysis and data-driven performance evaluation.
- Built a Random Forest model predicting cavity-receiver heat loss with <1% error on Nusselt number and deployed it as a web-based tool; designed HVAC systems and studied a transcritical CO₂ cycle (~8x COP gain via LSHE).

Graduate Research Assistant, IIT Bombay

Jul 2019 – Jul 2021

- Designed a high-temperature helical cavity receiver for concentrated solar power; ran 3D CFD simulations (ANSYS Fluent) of convective and radiative heat loss and derived Nusselt-number correlations.
- Built CFD-informed Random Forest surrogates with <1% Nusselt-number error and deployed them as a web-based engineering prediction tool.

Sales Engineer, Petrochem (Bangladesh) Limited

May 2018 – Jun 2019

- Designed, installed, and serviced solar PV and irrigation systems; led business development for 6 foreign companies.
- Led a team of 12 engineers and technicians, contributing to a 5% company revenue increase.

TEACHING

- Teaching Assistant, University of Arizona: AME 230 Thermodynamics (114-123 students) and AME 432 Heat Transfer (97 students); led review sessions and mentored over 350 undergraduates (2024-present).

SELECTED PUBLICATIONS

- Borshon, I. Z., et al. (2026). Evolution of grain boundaries in multi-principal alloys: insights into phase stability and structural complexity. *Computational Materials Science*, 268, 114628.
- Borshon, I. Z., et al. (2025). Deep learning analysis of solid-electrolyte interphase microstructures in lithium-ion batteries. *Advanced Materials Interfaces*, e00558.
- Borshon, I. Z., et al. (2024). Predicting column heights and elemental composition in experimental TEM images of high-entropy oxides using deep learning. *npj Computational Materials*, 10, 275.

HONORS & LEADERSHIP

- NSF Award No. 2311104; John Brammer Graduate Research Fellowship (OSU); MRS Best Poster Nominee (2025); DAAD Scholarship; Runner-up, CADRE Student Data Awards (Dell & Intel).
- President, Bangladesh Students' Association (200+ members), OSU; Treasurer, ASHRAE Student Chapter, OSU.